Continuity

Section **1.5**

A. Definition of Continuity

DEFINITION: A function f is continuous at a number a if

i. $f(a)$ exists

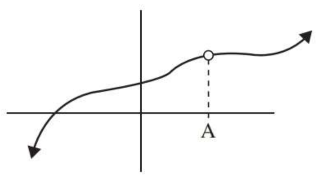
ii. $\lim_{x \to a} f(x)$ exists

iii. $\lim_{x \to a} f(x) = f(a)$

A function is defined as continuous only if it is continuous at every point in the domain of the function.

Examples: For each, determine whether the function is continuous (i.e., is $\lim_{x \to A} f(x) = f(A)$?)

1.



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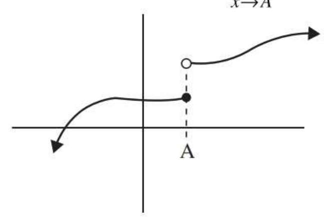

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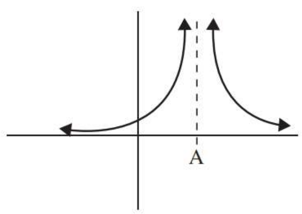

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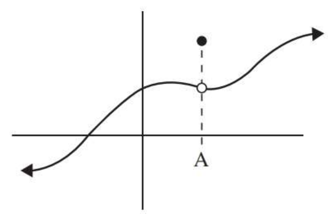


Examples: For each, determine whether the function is continuous. If not, where is the discontinuity?

5. **(video)** $f(x)=\dfrac{x^2-x-20}{x-5}$

6. **(video)** $f(x)=\begin{cases} -6-x & \text{if } x\le -3 \\ x & \text{if } -3<x\le 3 \\ (x-1)^2 & \text{if } x>3 \end{cases}$

7. **(video)** $h(x)=\begin{cases} x^2+1 & \textit{if } x\neq 2 \\ 5 & \textit{if } x=2 \end{cases}$

Examples: For each, determine the value of c so that function is continuous for all values of x.

8. **(video)** $f(x)=\begin{cases} cx^2+3x-7 & x\in(-\infty,3] \\ 2x+1 & x\in(3,\infty) \end{cases}$

9. $f(x)=\begin{cases} cx+7 & x\in(-\infty,8) \\ cx^2-7 & x\in[8,\infty) \end{cases}$

Where are these functions discontinuous?

10. $f(x)=\dfrac{\ln x+\sin x}{x^2-1}$

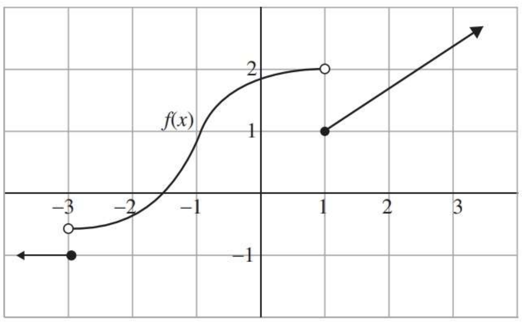
11. $f(x)=\ln(1+\cos x)$

DEFINITION: A function f is continuous from the RIGHT at a number a if $\lim_{x\to a^+} f(x) = f(a)$

A function f is continuous from the LEFT at a number a if $\lim_{x\to a^-} f(x) = f(a)$

Example

12.



f is continuous from the LEFT or RIGHT at

a. $x = -3$

b. $x = 1$

13. Show that $f(x)$ has a jump discontinuity at $x = 9$ by calculating the limits from the left and right at $x = 9$.

$$f(x) = \begin{cases} x^2 + 5x + 5 & \text{if } x < 9 \\ 14 & \text{if } x = 9 \\ -4x + 4 & \text{if } x > 9 \end{cases}$$

Theorem—If f and g are functions that are continuous at a number a, and c is a constant, then the following are also continuous at a:

i. $(f + g)$

ii. $(f - g)$

iii. $(f \cdot g)$

iv. $\left(\dfrac{f}{g}\right)$ if $g(a) \neq 0$

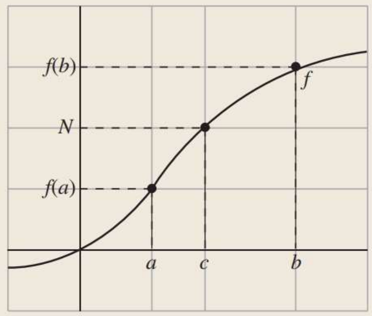
v. $c \cdot f$ or $c \cdot g$

Theorem—A **polynomial** function is continuous everywhere

A **rational** function is continuous everywhere it is defined

Theorem—Intermediate Value Theorem

If f is a function that is continuous on a closed interval $[a,b]$ where $f(a) \neq f(b)$ and N is a number such that $f(a) < N < f(b)$. Then there exist a number c such that $a < c < b$ and $f(c) = N$.



Examples

14. Show that $f(x) = x^2 - x - 2$ has a root on the interval $[1,3]$

15. Let f be a continuous function such that $f(1) < 0 < f(9)$. Then the Intermediate Value Theorem implies that $f(x) = 0$ on the interval (A, B). Give the values of A and B.